

Karthikheyaa Kurra

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EDUCATION

University of Delaware

M.S. Data Science, GPA 3.80/4.00

Newark, DE

Aug 2024 – May 2026

- **Coursework:** Probability Theory & Applications, Mathematical Techniques for Data Science (Optimization), Applied Multivariate Statistics, Machine Learning, Algorithm Design & Analysis, Research Software Engineering

Sathyabama Institute of Science & Technology

B.E. Computer Science & Engineering

Chennai, India

Oct 2020 – May 2024

EXPERIENCE

Research Associate, Scientific Machine Learning

Jun 2026 – Present

Department of Mathematical Sciences, University of Delaware (Advisor: Dr. Ke Chen)

Newark, DE

- Writing a paper on hybrid PDE solvers that combine classical methods and neural networks to solve a nonlinear elliptic PDE in a non-iterative manner through learned surrogates and by leveraging hierarchical structure.
- Own end-to-end research infrastructure: high-precision ground-truth generation via Chebyshev spectral discretizations and Newton iteration, automated dataset-validation suites, and reproducible PyTorch training and evaluation pipelines.
- Drove architecture and loss design through controlled ablation studies covering network normalization and physics-informed residual losses with adaptive loss balancing, alongside systematic wall-clock profiling of classical versus learned inference.
- Maintain a regression-locked, unit-tested codebase with automated experiment reports that set weekly research direction with the PI.

Research Assistant, Scientific Machine Learning

Jun 2025 – Jun 2026

Department of Mathematical Sciences, University of Delaware (Advisor: Dr. Ke Chen)

Newark, DE

- Joined Dr. Ke Chen's group to build neural surrogates for nonlinear PDE solvers, replacing per-instance Newton iteration with amortized single-pass inference on new problem instances.
- Established spectral ground-truth generation, dataset validation, and reproducible PyTorch experiment pipelines that underpin the project's training and evaluation workflow.

Research Intern, Computer Vision

Mar 2023 – Jun 2023

Indian Institute of Technology, Hyderabad

Hyderabad, India

- Case study applying supervised contrastive learning (Khosla et al., 2020) to facial expression recognition, comparing SupCon loss against a standard cross-entropy baseline in PyTorch with face detection and data augmentation.
- Analyzed learned embeddings with t-SNE to diagnose class separation issues and delivered a reproducible codebase to the lab.

Teaching Assistant, Machine Learning

Jan 2023 – Apr 2023

Sathyabama Institute of Science & Technology

Chennai, India

- Graded assignments and homework; prepared lecture slides and supplementary notes for the undergraduate machine learning course.
- Led review sessions on mathematics for machine learning, covering matrix algebra and multivariable calculus.

Summer Intern

Jun 2022 – Jul 2022

The University of Texas at Dallas

Dallas, TX

- Intensive summer program in machine learning and deep reinforcement learning: lectures, labs, and hands-on model building from first principles.

TECHNICAL SKILLS

Languages: Python, MATLAB, R, SQL, Bash, LaTeX

Scientific ML: PyTorch, NumPy/SciPy, Operator Learning, Physics-Informed Neural Networks (PINNs)

Tools: Git, Docker, Linux, HPC job scripting, VS Code, Cursor

READING GROUPS

Scientific Machine Learning Reading Group

2025 – Present

University of Delaware (Advisor: Dr. Ke Chen)

Newark, DE

- Weekly seminar with rotating student presentations of research literature; topics include Computational Optimal Transport (Peyré & Cuturi), Equivariant Neural Networks, and One-Shot Learning.